

Department of Physics and Astronomy

First Year

2024-25

Tutorial Sheet 6

PHYS10302 Vibrations and Waves

PHYS10342 Electricity and Magnetism

PHYS10372 Mathematics 2

This sheet covers tutorials in weeks 7 of semester 2.

Problem solving can be a difficult skill to master. If you are finding it tough going, it may be helpful to try the following strategy:

1. Identify what the problem is about. Try to visualise the problem. Drawing a diagram may help.
2. Identify clearly what you are being asked to find.
3. What information have you been given? Sometimes even the simple task of writing down this information can help.
4. What (physics) principles and equations can you use?
5. How do you put points 1, 2, 3 and 4 together to get a solution? What maths will you need?
6. It is usually best to try and solve the problem algebraically, using symbols, even if you are given numerical values. This enables you to check at each stage and at the end, that you have correct and consistent dimensions. The last thing to do is to put numbers in, if a numerical answer is required.
7. Is the answer reasonable? If the answer is numerical, is the order-of-magnitude appropriate? Does your algebraic answer behave in the correct way when the variables take on very large or very small values, or in other limits?

Dr Draga Pihler-Puzovic (Year Tutor)

draga.pihler-puzovic@manchester.ac.uk

PHYS10302 Vibrations and Waves

Week 7 Questions

1. Why does Helium make voices squeaky? You may use the fact that the velocity of sound is given by $v = \sqrt{\gamma P_0 / \rho}$, where P_0 is the equilibrium pressure, ρ is the mass per unit volume and γ is the adiabatic index.
2. A wave of amplitude 2 cm and wavelength 30 cm travels along a rope of density 200 g/m. At some point along the rope the density changes to 150 g/m. By calculating the refractive index $n = v/v'$, calculate the wavelength and amplitude of the transmitted wave. Is this answer reasonable?
3. If 1 Watt of light falls normally on a glass sheet of refractive index 1.4, what is the power reflected?
4. (a) Show that

$$\int_0^L dx \sin^2 \left(\frac{n\pi}{L} x \right) = \frac{L}{2},$$
$$\int_0^L dx \cos^2 \left(\frac{n\pi}{L} x \right) = \frac{L}{2},$$

where n is an integer.

- (b) In the lectures we derived an equation for the energy contained in a piece of string of linear mass density μ and length δx ,

$$E = \frac{1}{2} \mu \delta x \left[\left(\frac{\partial y}{\partial t} \right)^2 + v^2 \left(\frac{\partial y}{\partial x} \right)^2 \right],$$

where v denotes the wave velocity. If the string length is given by L , the standing wave solution for a single mode is,

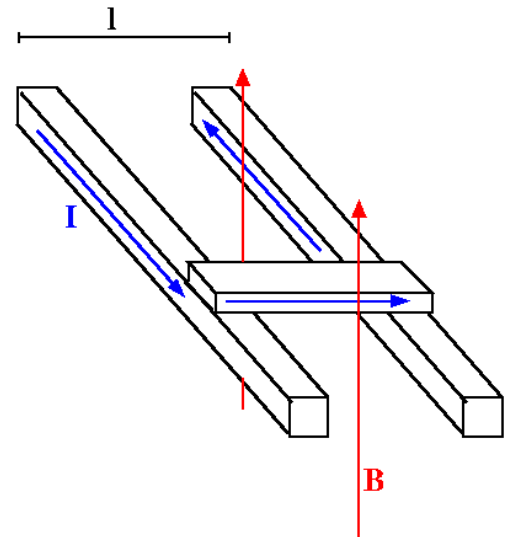
$$y_n(x, t) = A_n \sin \left(\frac{n\pi}{L} x \right) \cos \omega_n t,$$

with $\omega_n = v(n\pi/L)$. Calculate the energy E_n of the n^{th} mode in a string of length L . You may find the results from the first part of this question helpful. Comment on how E_n scales with the variables μ , L , A_n and ω_n .

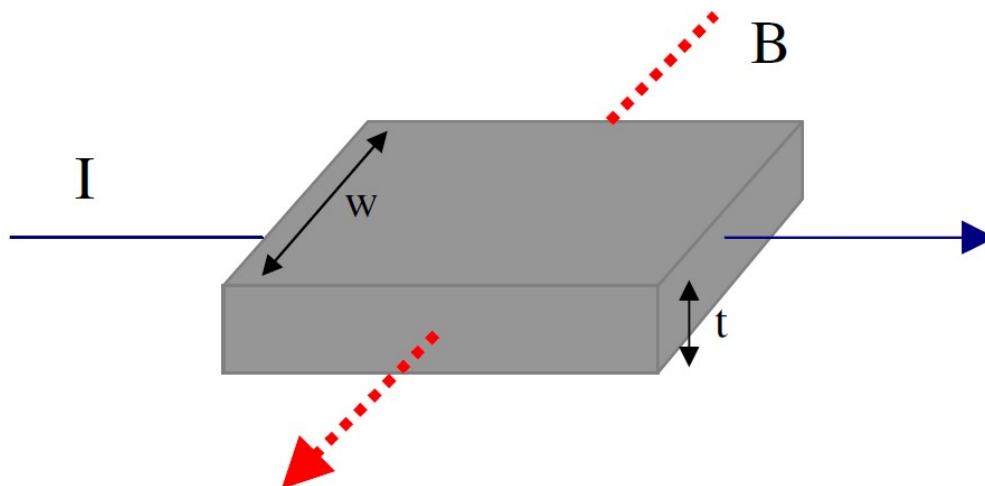
E&M Tutorial Sheet 6: Magnetic and Lorentz Forces

- 1) (Revision) Sketch non-zero fields that have: i) a net positive closed-surface flux; ii) zero circulation; iii) non-zero circulation.
- 2) A proton is travelling at a speed of $4.2 \times 10^6 \text{ ms}^{-1}$ in the xy plane in a region in which there is a constant magnetic field 0.31 T in the z direction. Calculate the radius of curvature of the proton's trajectory.

- 3) In an electromagnetic 'railgun', a conducting bar of mass m and length d slides along rails that are connected to a voltage source. The bar completes the circuit allowing a current I to flow. A vertical magnetic field fills the region between the rails, as shown in the diagram. Assuming the current and magnetic field are constant



- a) Find the magnitude and direction of the net force on the conducting bar. Neglect friction, air resistance and electrical resistance.
- b) Find the velocity v of the bar after it has travelled a distance s along the rails, starting at rest.
- c) If $B = 4.0 \text{ T}$, $I = 2.0 \times 10^6 \text{ A}$, $m = 2 \text{ kg}$ and $d = 10 \text{ cm}$, calculate how long the rails would need to be to achieve exit speeds of (i) 3 km s^{-1} (the world record) and (ii) 11.2 km s^{-1} (Earth's escape velocity).
- 4) A current flows through a rectangular slab of conducting material, in the presence of a uniform magnetic field, as shown in the diagram below.



- a) If the moving charges are positively charged, in which direction are they deflected by the magnetic field? This deflection causes an accumulation of charge on the upper or lower surface of the slab. This builds up an electric field, which eventually produces an electric force that counterbalances the magnetic deflection. Equilibrium is set up when the two exactly cancel and this is known as the Hall effect, discovered by Erwin Hall in 1897 when he was a Ph.D. student.

- b) Through this mechanism there is a potential difference setup between the top and bottom of the slab called the Hall voltage. Find an expression for the Hall voltage in terms of B , the dimensions of the bar and v , the velocity of the charge carriers.
- c) How would things change if the charge carriers were negative instead?
- d) Find an expression for the current, I , in terms of the number of charge carriers per unit volume, n , the drift velocity, v , the charge they carry, q , and the cross sectional area through which the current flows, A .
- e) In a Hall effect measurement of a piece of sodium metal, a Hall field of $25 \mu\text{V cm}^{-1}$ was obtained for a current density of 1000 A cm^{-2} in a transverse field of 1 T . Find the density of charge carriers. Compare it to the number of atoms per unit volume. The density of sodium is 0.97 g cm^{-3} and it has an atomic mass of 23.0 .

PHYS10372 Mathematics 2: Tutorial-06 Questions

These questions cover the material from week-6 of PHYS10372: calculating circulation and Stokes' theorem.

1. Use Stokes' theorem to evaluate the following line integrals. You might find it helpful to sketch the surfaces, ensuring that you get the loops that describe the intersections correct.

(a) The anticlockwise circulation of the vector field: $\mathbf{A} = 5x\hat{\mathbf{j}}$ around the circle $x^2 + y^2 = a^2$ and $z = 0$.

[Answer: $5\pi a^2$]

(b) The anticlockwise circulation of the vector field: $\mathbf{A} = z^2\hat{\mathbf{i}} - y^2\hat{\mathbf{j}}$ around a square of side a lying in the xz plane, two of whose edges lie along the x and z axes

[Answer: a^3]

(c) $\mathbf{A} = 4y\hat{\mathbf{i}} + 2z\hat{\mathbf{j}} + 6y\hat{\mathbf{k}}$ around the intersection of the ellipsoid $x^2 + y^2 + \frac{z^2}{4} = 1$ and the plane $z = 1$ in the anticlockwise direction as viewed from the origin.

[Answer: $+3\pi$]

(d) $\mathbf{A} = -2z\hat{\mathbf{i}} + x\hat{\mathbf{j}} - x\hat{\mathbf{k}}$ around the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = y + 1$ in the clockwise direction as viewed from the origin.

[Answer: $+2\pi$]

(e) $\mathbf{A} = y\hat{\mathbf{i}} + xz^3\hat{\mathbf{j}} - zy^3\hat{\mathbf{k}}$ around the circle $x^2 + y^2 = a^2$ when $z = b (< 0)$ in the anticlockwise direction as viewed from the origin.

[Answer: $\pi a^2(b^3 - 1)$]

2. Evaluate the line integral of the vector field $\mathbf{A} = \frac{1}{x^2+y^2}(-y\hat{\mathbf{i}} + x\hat{\mathbf{j}})$ anti-clockwise around the circle $x^2 + y^2 = 1$ at $z = 0$.

[Answer: 2π]

This is most easily solved by converting to plane polar coordinates